

Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	Flood Prediction System
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Used to create a responsive and user-friendly web interface for entering weather parameters and displaying flood prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, loads the trained machine learning model, processes user inputs, and returns prediction results efficiently.
3	Database / Data Storage	Excel Dataset (XLSX), Joblib (.save)	Historical weather data is stored in an Excel dataset for model training, while the trained model and scaler are saved using Joblib for fast loading during prediction.
4	Cloud / Hosting / Deployment	Local Flask Development Server	The application is deployed locally for development and testing, providing a simple environment for demonstrating the project.

5	Version Control & CI/CD	Git, GitHub	Git is used for version control, while GitHub stores the project repository and documentation for collaboration and submission.
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib	These libraries are used for data preprocessing, visualization, model training, evaluation, and saving/loading the trained machine learning model.